



وزارة الاتصالات  
وتكنولوجيا المعلومات



# ANUBIS X

A Threat-First Intelligent Surveillance System



**Program:**

Digilians Nine-Month Professional Diploma  
in Applied AI & Data Analytics



**Project:**

Graduation Project



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# Agenda



Introduction



Problem  
Definition



Motivation &  
Objectives



Methodology



System  
Architecture



Implementation



Results &  
Analysis



Future Work



References



Thank You /  
Q&A

# Introduction



## 01 Many CCTV Feeds



Security teams monitor numerous cameras simultaneously, making it hard to catch critical events.

## 02 Limited Human Attention



Human attention is limited, leading to delayed response and missed threats.

## 03 Real Time-Critical Threat



Threats happen in real time. Every second of delay can have serious consequences.

## ANUBIS X

A Threat-First Intelligent Surveillance System



Threat  
Detection



Associated  
Person



Reviewable  
Evidence

ANUBIS X detects threats first, associates them with the relevant person, and provides reviewable evidence for faster, smarter decisions.



# Problem Definition

## Why Traditional CCTV Falls Short



### 01 Many Feeds



Too many cameras.  
Too much to watch.

### 02 Human Overload



Hours of monitoring.  
Fatigue sets in.

### 03 Missed Critical Events



Critical moments blend in.  
Nothing is flagged.

### 04 Delayed Response



Response starts late.  
Impact is already done.

## Why AI Is Still Difficult



Small  
Weapons



Occlusion



Poor  
Lighting



Look-Alike  
Objects



Passive CCTV records video, but **does not reliably** surface what matters in time.

# Motivation & Objectives



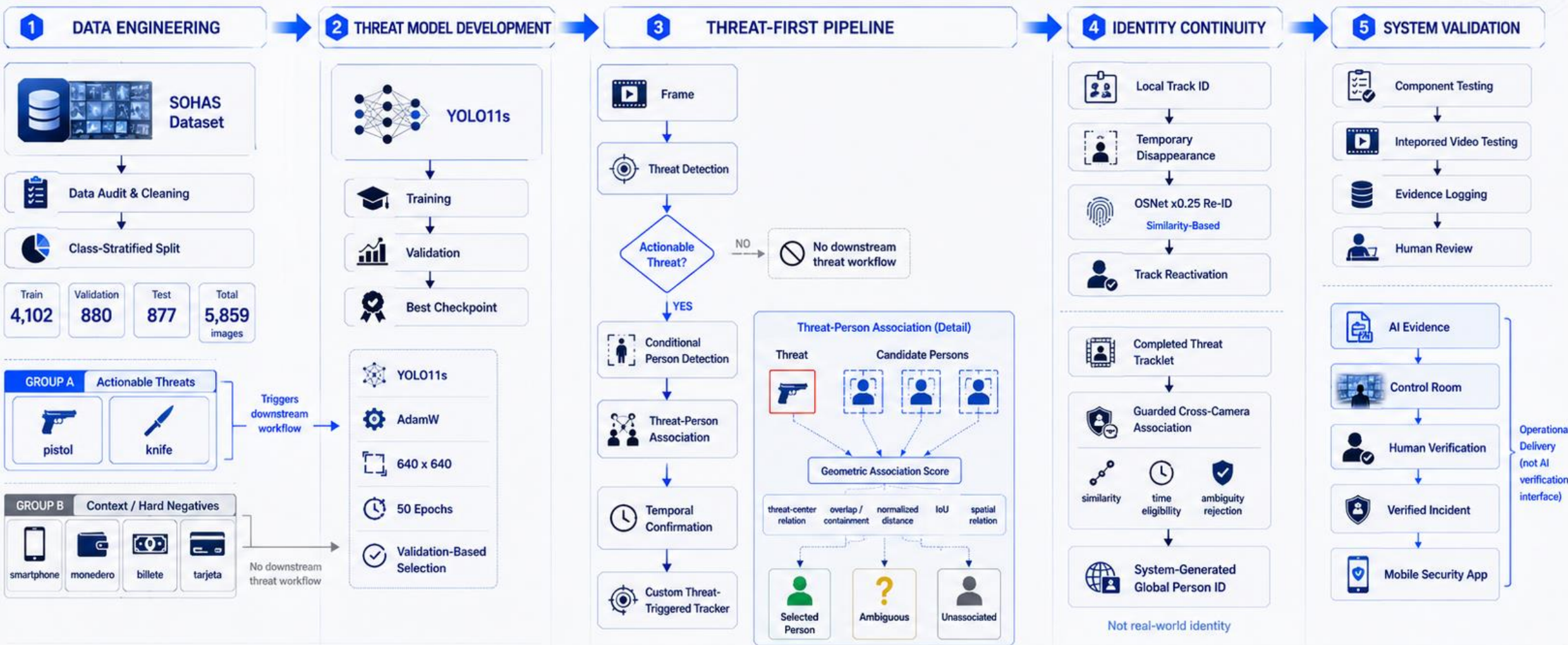
## OPERATIONAL FLOW



 **ANUBIS X** is designed to surface actionable threats, connect them to the relevant person, and deliver verified evidence for response.

# METHODOLOGY

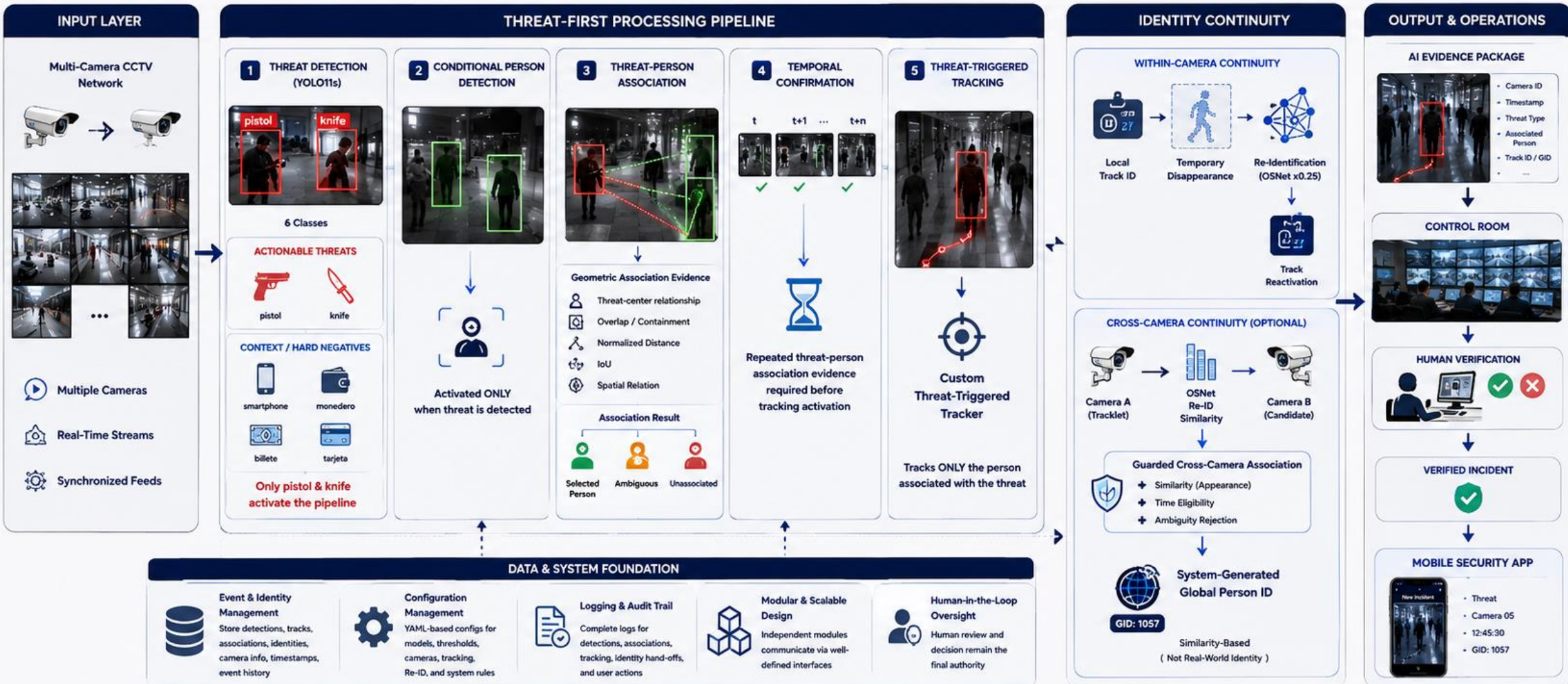
From Data Preparation to Operational Validation



# ARCHITECTURE

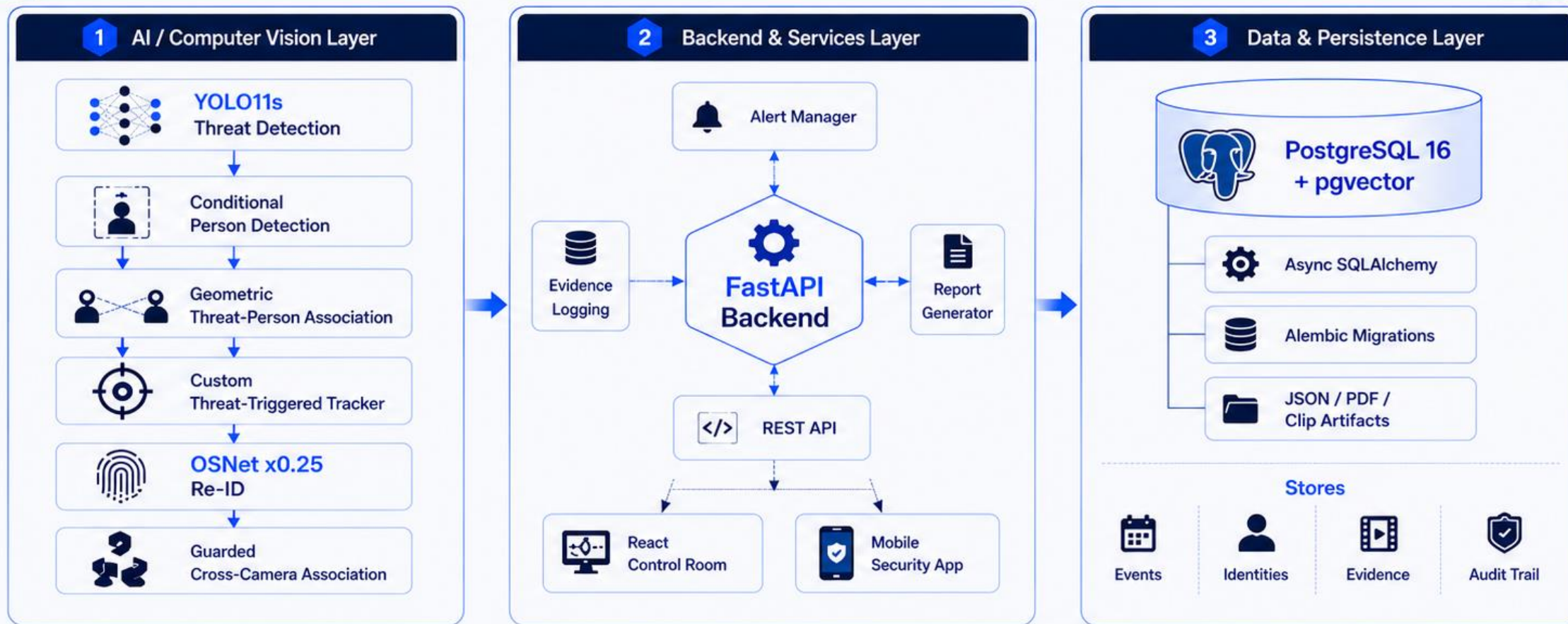


## ANUBIS X – Threat-First Intelligent Surveillance System



# IMPLEMENTATION

## Core System Layers



Deployment / Environment



Docker Compose



Nginx



Windows



Python 3.10.11



PyTorch 2.6.0+cu124



RTX 2000 Ada

# IMPLEMENTATION

## Operational Delivery Layers (Part 2)



### 4 Control Room Layer



#### Core Purpose

Real-time monitoring, human verification, and evidence management.

#### Key Capabilities

- Live Monitoring
- Event Logs
- Threat Clips
- Incident Reports

#### Tools Used



React  
(Vite)



Tailwind  
CSS



WebSocket  
(STOMP)



REST API  
(FastAPI)



### 5 Mobile Response Layer



#### Core Purpose

Deliver verified incidents and evidence to field units securely and in real time.

#### Key Capabilities

- Incident Inbox
- Evidence Viewer
- Notifications
- Response Status

#### Tools Used



React Native



JWT  
Authentication



Firebase  
Cloud Messaging



HTTPS  
(Secure API)

# RESULTS



## THREAT DETECTION



**86.3%**  
Precision



**89.8%**  
Recall



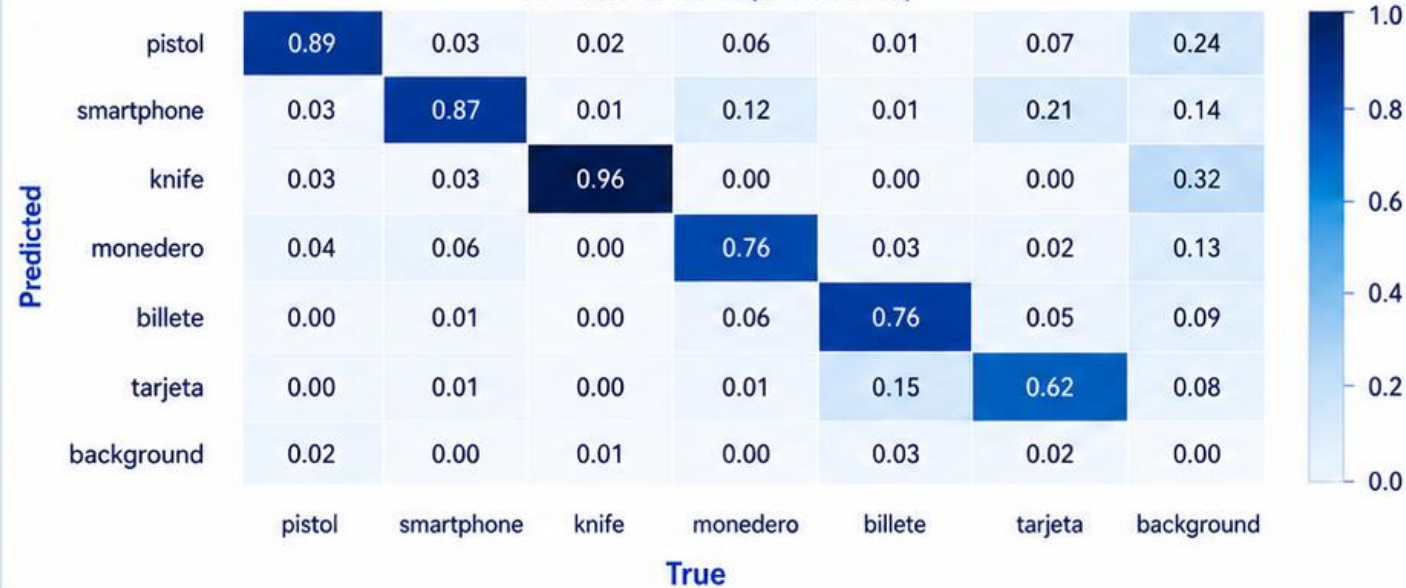
**94.1%**  
mAP@0.5



**79.2%**  
mAP@0.5:0.95

## DETAILED MODEL OUTPUTS

Confusion Matrix (Normalized)



## OSNet Re-ID



Rank-1

**48.3%**



Rank-5

**71.5%**



mAP

**59.5%**



Appearance-based similarity

# FUTURE WORK



Building the next generation of threat-first intelligent surveillance.

01

## Stronger & Smarter Models



Collect more real CCTV data and improve detection, association, tracking, and Re-ID performance.

02

## Multi-Camera Scale & Continuity



Enhance cross-camera linking and long-term identity continuity in large-scale networks.

03

## Robust in Real-World Conditions



Improve robustness in crowded, occluded, low-light, and adverse weather scenarios.

04

## System Performance & Scalability



Optimize speed, reduce latency, and ensure scalability for larger deployments.

05

## Operational & User Experience



Expand the mobile security app with real-time alerts, location services, and richer incident context.

## Our Vision

A reliable, real-time, and human-centered surveillance ecosystem that enhances situational awareness and public safety.



# REFERENCES

Key works that informed the design of key components within the ANUBIS X pipeline.

01



## Ultralytics YOLO11

Ultralytics  
Official Documentation, 2024

Threat Detection & Person Detection

02



## OSNet: Omni-Scale Feature Learning for Person Re-Identification

K. Zhou et al.  
ICCV, 2019

Re-ID

03



## Comparative Analysis of Previous YOLO Detectors and YOLOv26s for Real-Time Weapon Detection in Video Surveillance

C. J. Fierro Silva, C. Del-Valle-Soto, C. Bran, J. Varela-Aldás  
Frontiers in Computer Science, 2026

Weapon Detection Benchmark

04



## Improving Armed People Detection in Video Surveillance Systems by Combining Object Detection and Pose Estimation

Amado-Garfias et al.  
2024

Threat-Person Association

05



## Multi-Camera Multi-Object Tracking: A Review

Amosa et al.  
2023

Identity Continuity

06



## Multi-Camera People Tracking

Nguyen et al.  
2023

Cross-Camera Association

These works informed the design of key components within the **ANUBIS X** pipeline.



ANUBIS X

THANK YOU